



Cold sintering for in-space materials fabrication: a science, engineering, and future directions

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Abstract

The Cold Sintering Process (CSP) has recently emerged as a transformative approach for densifying ceramics and composites at temperatures below 300 °C using pressure-assisted dissolution–precipitation. This low-energy route is particularly compelling for in-situ resource utilization (ISRU) in extraterrestrial environments, where transporting construction materials from Earth is impractical. This review critically examines the potential of CSP for fabricating structural and functional materials in space, spanning lunar, Martian, orbital, deep-space, and asteroid contexts. We analyze the fundamental mechanisms of CSP, recent experimental demonstrations on lunar and Martian regolith simulants, and its integration with additive manufacturing, robotics, and artificial intelligence. Case studies highlight the production of high-strength regolith bricks (> 200 MPa) at low energy cost, showing CSP's feasibility for building load-bearing structures on the Moon and Mars. We further explore frontier applications, including radiation shielding, multifunctional composites, and autonomous construction systems. Challenges such as solvent sourcing, high-pressure requirements, and environmental constraints are discussed in detail. Finally, we outline research directions toward scaling CSP for large habitats, integrating it with space architecture, and deploying it as part of closed-loop resource systems. CSP is positioned as a cornerstone technology for sustainable space infrastructure, enabling humankind to build resilient habitats and functional systems beyond Earth.

Keywords Cold sintering process · In-situ resource utilization (ISRU) · Lunar regolith · Martian regolith · Space manufacturing · Additive manufacturing

1 Introduction

The vision of sustained human presence beyond Earth demands revolutionary approaches to construction and manufacturing using local resources (Sun et al. 2024). In particular, in-situ resource utilization (ISRU) on the Moon, Mars, and in free space offers a way to fabricate habitats and infrastructure without transporting all materials from Earth (Zamani et al. 2024). Traditional sintering of ceramics and glass requires very high temperatures (often > 1000 °C), which is energy-intensive and challenging in off-world environments (Vakifahmetoglu and Karacasulu 2020). The cold

sintering process (CSP) has emerged as a low-temperature alternative that can densify ceramic powders into solid parts by combining moderate heat (often below ~ 300 °C), high pressure, and a transient liquid phase (Rahimi Mamaghani and Parvin 2025a; Evangeline and Annamalai 2025). In CSP, a small amount of solvent (often water or a water-based solution) is introduced to the powder compact; under pressure and gentle heating, the solvent partially dissolves the particle surfaces, allowing material transport and reprecipitation at contact points (Grasso et al. 2020). This dissolution–precipitation mechanism enables densification of the material without the need for full melting. Because CSP replaces high-temperature diffusion with pressure-assisted dissolution–precipitation, it can deliver comparable densification at a fraction of the thermal energy budget, making it attractive for power-constrained missions (Ndayishimiye et al. 2020a).

Researchers have shown CSP can densify a broad range of ceramics, including oxide electrolytes, piezoelectrics, and electronic materials (Murutoglu et al. 2022; Zubairi

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et al. 2023). The process is also amenable to composites: polymers, metals, or glass phases can be co-sintered with the ceramic powder, producing multi-material parts in one step (Lai et al. 2025). For example, Hu et al. (2024) showed that polyester polymers such as PLA and PCL can directly assist the densification of insoluble ceramics (e.g., Hydroxyapatite and Al_2O_3) without any liquid additives, achieving near-theoretical densities below 100 °C. Complementary to this approach, Nie et al. (2025) showed that CSP can regulate polymer–ceramic interfaces in solid-state electrolytes, where a poly(ionic liquid) gel phase uniformly distributed at lithium aluminum titanium phosphate (LATP) grain boundaries reduced interfacial resistance and enabled high ionic conductivity (0.42 mS cm^{-1} at room temperature). These results highlight the versatility of CSP in integrating polymers with ceramics, either as structural binders or as functional conductive interphases. These multimaterial possibilities are particularly attractive in space, where complex component integration can reduce assembly steps.

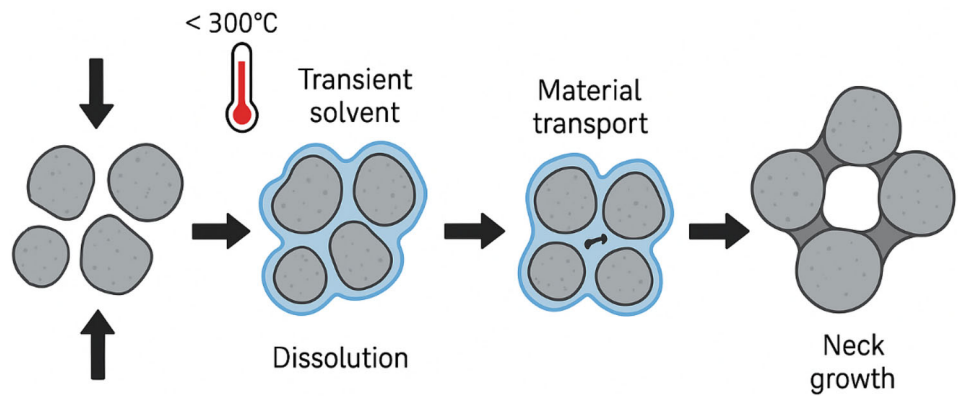
Importantly, CSP's low processing temperature translates to dramatic energy savings and compatibility with heat-sensitive components. Conventional sintering might consume on the order of 5–10 kWh per kilogram of material, whereas Liu et al. reported that CSP densifying lunar regolith consumed only $\sim 0.86 \text{ kWh}$ under laboratory conditions (Guo et al. 2024). This denotes an approximately 6.5-fold decrease in energy, which is highly significant for space missions, where power is limited. Furthermore, avoiding high heat means less stringent thermal shielding and less stress on equipment. As most recently, the absence of high-temperature requirements in CSP facilitates low-energy, sustainable manufacturing and production of thermal-sensitive materials, such as polymer composites (Si et al. 2025), nano-scaled (Zhang et al. 2024a) and quantum ceramics (Mamaghani and Parvin 2025). In summary, the CSP process leverages a transient solvent and applies pressure to dissolve and reform particle contacts at low temperature. The result is dense ceramic parts with strong bonding but minimal thermal budget. This unique mechanism of pressure-driven dissolution–precipitation underpins CSP's promise for space fabrication.

Recently, Liu et al. demonstrated that lunar regolith simulant can be cold-sintered into dense, high-strength bricks that retain mechanical integrity under extreme thermal cycling, while consuming only $\sim 0.86 \text{ kWh}$ of energy, over six times lower than conventional high-temperature sintering (Liu et al. 2024). Early CSP research focused on electronic ceramics (Seo et al. 2017), such as LiFePO_4 and BaTiO_3 , but the technique is now being applied to space-architectural materials. The CSP's compatibility with a variety of powders (including mixed compositions) and its low-temperature operation make it attractive for off-Earth fabrication (Ding et al. 2024).

By processing local regolith (the loose soil on planetary surfaces or asteroids) into structural building blocks (Ellery 2022), CSP could enable rapid construction of habitats, radiation shields, roadways, and even components such as heat exchangers or optical elements without costly and risky transports. Moreover, CSP naturally integrates with additive manufacturing (AM) and composite materials approaches (Marín-Cortés et al. 2024). Herein, a printed regolith layer can be cold-sintered in place to create a monolithic wall, or regolith particles can be co-sintered with polymers or metals to form functionally graded materials (Suhaizan et al. 2024). The integration of CSP with AM is viable for producing complex geometries, not as a stand-alone step for finished parts, but as a complementary post-processing or localized consolidation method (Sinha et al. 2021). The AM can fabricate near-net-shape ceramic green bodies with designed porosity, which serve as scaffolds for subsequent densification via conformal pressing or incremental compression (Baux et al. 2021). Alternatively, pressure can be applied locally during or after material deposition, enabling controlled densification that preserves intricate designs. This hybrid AM-CSP strategy decouples shape formation from final material strengthening. It maintains the design freedom critical for in-space manufacturing, such as creating complex internal channels or lightweight lattices (Paek et al. 2022), while leveraging CSP to achieve the required mechanical performance from consolidated in-situ resources.

In this article, we review the state of CSP as a tool for in-space fabrication. We begin by explaining the fundamental mechanisms of CSP, then survey its applications across extraterrestrial environments (Moon, Mars, orbit, deep space, and asteroids). We highlight key experimental case studies (such as cold-sintered lunar and Martian simulants) to illustrate current capabilities. The challenges of implementing CSP in space, from resource supply to microgravity effects, are discussed in a dedicated section. We also examine how CSP could integrate with emerging technologies in space architecture, robotics, and artificial intelligence (AI). Finally, we suggest future research directions and explore the role CSP may play in building off-Earth infrastructure and habitats. Throughout, we emphasize recent literature and the multidisciplinary nature of this field, aiming to provide a deep, forward-looking synthesis for both scientists and engineers. This review places CSP within current ISRU and exploration roadmaps (e.g., lunar Artemis objectives and planned cislunar infrastructure), arguing that low-temperature, low-energy fabrication techniques are directly relevant to near-term demonstration missions and long-term habitat architectures.

Fig. 1 Schematic illustration of the CSP mechanism. Under applied pressure and moderate heat ($< 300\text{ }^{\circ}\text{C}$), a transient liquid dissolves particle surfaces and reprecipitates at contact points, densifying the compact without the need for high temperatures



2 Mechanisms of the cold sintering process

The CSP is fundamentally a pressure-assisted, low-temperature densification process (Guo et al. 2016a). Unlike traditional sintering, which relies on high thermal energy to induce surface diffusion and grain growth (Wang et al. 2025), CSP employs a transient liquid or solvated phase to transport material between particles under moderate conditions (Rahimi Mamaghani and Parvin 2025a; Guo et al. 2016b). In a typical CSP experiment, a finely powdered ceramic or composite (often $< 100\text{ }\mu\text{m}$) is mixed with a small amount of solvent or solution (such as water, acidic, or basic solutions) and then subjected to uniaxial pressure (tens to hundreds of MPa) at an elevated but still modest temperature (often $100\text{--}300\text{ }^{\circ}\text{C}$). The solvent wets the particle surfaces and partially dissolves their constituents. Under the applied pressure, the particles rearrange, and dissolved species migrate through the liquid phase to zones of contact or high stress. When the solvent evaporates or is removed (e.g., by continued heating or by washing), the dissolved material reprecipitates at the grain contacts, forming bridging “necks” that bind the particles. The overall effect is densification: pores shrink and the compact’s porosity can fall dramatically, often to relative densities of 80–90% or higher (Galotta and Sglavo 2021; Guo et al. 2016c). For space deployment this implies two practical constraints that must be engineered: (1) feedstock often needs comminution to fine powders (typically $< 100\text{ }\mu\text{m}$) to enable efficient dissolution–reprecipitation and packing, and (2) CSP commonly requires sustained uniaxial or isostatic pressures in the order of $100\text{--}300\text{ MPa}$, which drives the design of compact, low-mass press hardware for off-Earth use. The Fraunhofer IKTS research group summarizes this as a dissolution–precipitation mechanism (Sharipova et al. 2025). With the aid of the liquid phase, the solid particles are rearranged under moderate temperatures and high pressures. The diffusive mass transport is followed by the precipitation process, where amorphous phases form at the grain boundaries. After the solvent has evaporated, these

phases can either remain amorphous or recrystallize, depending on the material properties (Ndayishimiye et al. 2020a). In essence, CSP temporarily creates a local “sintering liquid” that allows atoms or ions to relocate at much lower overall heat input than traditional processes. The fundamental stages of CSP, containing particle dissolution, reprecipitation, and densification, are schematically illustrated in Fig. 1. Key requirements include partial solubility of the material in the chosen solvent and high applied stress to drive the pressure-solution creep. Materials with ionic bonding, reactive surfaces, or low lattice energy tend to sinter well under CSP (Grasso et al. 2020).

Beyond case-based demonstrations, the CSP mechanism is governed by coupled effects of solvent chemistry, particle characteristics, and processing conditions (Evangeline and Annamalai 2025; Rahimi Mamaghani et al. 2025b). The choice of solvent critically controls the dissolution kinetics of surface species and the formation of transient intergranular liquids, thereby influencing phase assemblage and bonding pathways (Ndayishimiye et al. 2020a). Aqueous or mildly ionic solvents promote pressure-assisted dissolution–precipitation and the formation of amorphous or nanocrystalline interphases at grain boundaries, whereas less reactive solvents suppress mass transport and limit densification (Rahimi Mamaghani et al. 2025a). For example, increasing the concentration of a sodium hydroxide (NaOH) solution significantly enhances the relative density and mechanical properties of sintered compacts (Karacasulu et al. 2023). Particle size and surface chemistry further modulate CSP efficiency, with finer particles, higher defect densities, and partially amorphous surfaces exhibiting enhanced solubility and faster interfacial reprecipitation. Increasing pressure accelerates these processes by increasing chemical potential gradients, while moderate temperature (typically below $300\text{ }^{\circ}\text{C}$) primarily governs solvent mobility and evaporation rather than bulk diffusion. As a result, CSP commonly yields dense materials with amorphous or partially recrystallized grain-boundary phases, distinct from the equilibrium microstructures produced by high-temperature sintering.

This process can produce materials with high compressive strength, demonstrating its effectiveness despite the low temperatures.

2.1 Comparative positioning of cold sintering within space manufacturing technologies

The deployment of material processing technologies for space ISRU is inherently constrained by energy availability, system mass, operational reliability, and environmental extremes. While CSP offers unique advantages for low-temperature consolidation, it represents only one of several emerging or established space manufacturing approaches. To avoid a one-sided assessment and to clarify the differentiated positioning of CSP within the broader space manufacturing technology ecosystem, a comparative technology framework is presented here, benchmarking CSP against mainstream alternatives including microwave sintering, solar sintering, and geopolymer molding. From an energy consumption perspective, CSP is distinguished by its ability to achieve densification at temperatures far below those required for conventional or radiative sintering techniques, reducing required temperatures from ~ 1400 to $3000\text{ }^{\circ}\text{C}$ to approximately $300\text{ }^{\circ}\text{C}$ (Evangeline and Annamalai 2025). Microwave and solar sintering rely on sustained high-temperature exposure, demanding either significant electrical power or large-area optical concentrators. In contrast, CSP shifts the primary energy input from thermal to mechanical domains, enabling consolidation under moderate pressure and low thermal budgets. However, this advantage is balanced by the need for pressure-generation systems, which introduce their own engineering challenges. Notably, hybrid processes like the Microwave Cold Sintering Process (MW-CSP) have been developed to densify ceramics under pressureless conditions, merging the benefits of microwave resonance and dissolution–precipitation phenomena to dramatically reduce both time and energy consumption (Li et al. 2025a).

In terms of equipment complexity and system mass, CSP-based systems can be comparatively compact, particularly when designed for modular or batch-scale operation. Microwave sintering requires waveguides, shielding, and precise electromagnetic control, while solar sintering depends on large concentrators, precise tracking mechanisms, and line-of-sight exposure. Geopolymer molding generally involves lower thermal complexity but requires extensive precursor processing, curing time, and chemical handling (Lee et al. 2022). The CSP occupies an intermediate position, combining moderate mechanical complexity with simplified thermal management. Raw material compatibility represents a key dimension differentiating these technologies. High-temperature sintering approaches are well suited to oxide-rich regolith but may be sensitive to

compositional heterogeneity and volatile loss. Geopolymer systems rely heavily on specific aluminosilicate chemistry and the availability of alkaline activators like NaOH or KOH (Mendoza-Cachú et al. 2024). The CSP, by contrast, exhibits flexibility toward mixed or fine-grained oxide and silicate feedstocks, provided that partial solubility or surface reactivity can be activated through appropriate solvent selection. This makes CSP particularly attractive for heterogeneous regolith-derived materials, while simultaneously limiting its applicability to fully metallic or insoluble phases.

Regarding achievable mechanical properties, high-temperature sintering methods generally yield superior densification and strength, particularly for load-bearing structural elements (Rahimi Mamaghani and Parvin 2023). The CSP-fabricated materials, while capable of achieving useful mechanical integrity, are more naturally suited to functional components, radiation shielding, bonding layers, and modular structural elements rather than primary load-bearing infrastructure. However, materials sintered via advanced methods like MW-CSP have demonstrated mechanical and dielectric properties improved by 50–95% over other pressureless technologies, highlighting the potential for performance enhancement (Li et al. 2025a). This distinction underscores CSP's role as a complementary, rather than competing, technology within space construction strategies. Finally, environmental robustness varies across technologies. Solar sintering is inherently constrained by illumination conditions and dust contamination, while microwave systems must contend with electromagnetic interference and coupling efficiency. The CSP mechanisms are largely independent of ambient atmosphere and gravity, but require careful management of pressure delivery and solvent containment under vacuum conditions. These trade-offs underscore the importance of context-specific technology selection rather than universal solutions. A comparative summary of these technologies across key performance dimensions is provided in Table 1. Collectively, this framework positions CSP as a low-energy, material-flexible, and mechanically moderate processing route that complements, rather than replaces, high-temperature or chemically driven space manufacturing technologies.

3 Applications in extraterrestrial environments

While experimental validation of CSP has so far been demonstrated primarily for lunar and Martian basaltic regolith simulants, the potential applicability of CSP to other extraterrestrial materials does not rely on planetary origin per se, but on a set of mechanistic criteria intrinsic to the CSP. The CSP is fundamentally governed by pressure-assisted dissolution–precipitation, particle rearrangement under stress,

Table 1 Comparative framework for space iSRU manufacturing technologies

Technology	Energy demand	Equipment complexity	Raw material compatibility	Mechanical performance	Key limitations
Cold sintering process (CSP)	Low thermal, moderate mechanical energy (typically ≤ 300 °C compared to 1400–3000 °C for conventional sintering)	Mechanical press with solvent delivery and containment systems	Oxide- and silicate-rich, fine-grained materials; requires partial solubility or surface reactivity in transient liquids	Moderate to high for optimized variants (e.g., microwave-assisted CSP)	Solvent management under vacuum; pressure application and reaction-force control in reduced gravity
Microwave sintering	High electrical energy demand due to volumetric heating at elevated temperatures	High (microwave generators, waveguides, shielding, electromagnetic control)	Dielectric or semi-dielectric materials; sensitive to composition and coupling efficiency	High (superior densification and strength achievable)	Power-intensive; requires precise electromagnetic tuning and shielding
Solar sintering	High optical energy input via concentrated solar radiation	High (large solar concentrators, optical tracking, alignment systems)	Oxide-rich regolith materials	High (demonstrated for lunar regolith analogs)	Strongly dependent on illumination conditions; dust contamination; limited operation windows
Geopolymer molding	Low to moderate (ambient or low-temperature curing)	Chemical processing systems (mixing, molding, safe handling of alkaline activators)	Aluminosilicate-rich materials only; requires specific chemistry and alkaline activators (e.g., NaOH, KOH)	Moderate (compressive strength commonly reaching ≥ 100 MPa)	Long curing times; dependence on activator availability and handling safety

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and transient liquid-mediated mass transport (Guo et al. 2016a). Therefore, materials that exhibit partial solubility in an available solvent, reactive or defect-rich surfaces, and sufficient fine particulate fractions can, in principle, be consolidated via CSP irrespective of their celestial provenance (Funahashi et al. 2017). In this context, references to asteroid or deep-space materials in this review should be understood as conditional applicability scenarios, rather than experimentally validated cases. From a mechanistic perspective, CSP applicability requires: (i) Partial chemical solubility of one or more regolith phases in a transient liquid (e.g., water, alkaline solutions, or low-volatility solvents), (ii) Fine particle size distributions enabling high contact area and pressure-solution creep, (iii) Pressure-assisted mass transport, independent of gravity, and (iv) Thermodynamic stability of precipitated phases at low temperature (Lai et al. 2025; Aiyeki et al. 2025).

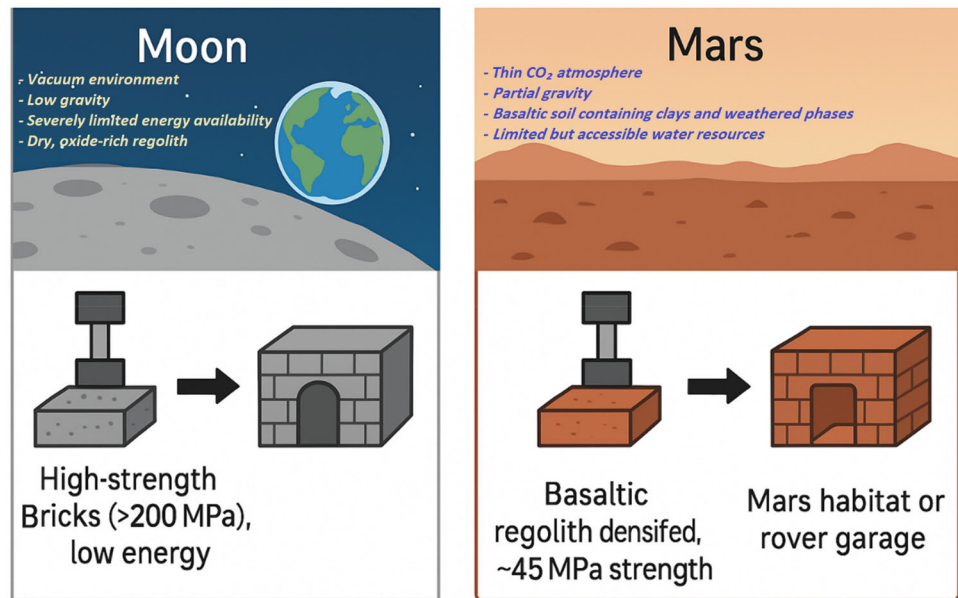
The CSP's unique features (low energy, rapid processing, and feedstock flexibility) suggest many possible applications off Earth. The environmental context (gravity, atmosphere, temperature extremes) varies greatly between the Moon, Mars, orbital stations, deep space, and asteroid surfaces. To

contextualize the potential of CSP beyond Earth, Fig. 2 conceptually illustrates its application in fabricating structural components from lunar and Martian regolith. Here we discuss how CSP might be deployed in each of these settings, highlighting relevant factors and opportunities.

3.1 Mechanistic criteria governing CSP applicability across extraterrestrial materials

The applicability of the CSP to extraterrestrial materials is fundamentally governed by material-specific chemical and transport mechanisms, rather than by the planetary origin of the feedstock. Although experimental demonstrations of CSP have been primarily conducted on terrestrial ceramics and lunar or Martian basaltic regolith simulants, the extension of CSP concepts to other extraterrestrial environments requires a mechanistic, criteria-based assessment, rather than direct extrapolation of existing experimental data. The CSP is widely understood as a pressure-assisted, liquid-mediated densification process, involving transient dissolution of surface species, particle rearrangement under applied stress, and subsequent precipitation or solid-state bonding at temperatures far below those required for conventional sintering

Fig. 2 Conceptual applications of CSP in extraterrestrial environments. On the Moon, CSP can produce high-strength regolith bricks (> 200 MPa) with low energy consumption. On Mars, CSP enables densification of basaltic soil into structural components, complementing ISRU-based habitat construction



(Galotta and Sglavo 2021; Guo et al. 2016c). Therefore, the feasibility of CSP for a given extraterrestrial material depends on whether its constituent phases can participate in these mechanisms under accessible processing conditions.

3.1.1 Partial chemical solubility or reactivity in a transient liquid phase

A prerequisite for CSP densification is the presence of at least partial solubility or surface reactivity of one or more material phases in a transient liquid environment. In oxide- and silicate-rich systems, such as lunar and Martian basaltic regolith simulants, this requirement is readily satisfied through alkali-assisted dissolution–precipitation pathways, which have been experimentally demonstrated to enable significant densification at low temperatures (Xue and Qiao 2025a; Linke et al. 2020). In contrast, extraterrestrial materials enriched in metallic phases, sulfides, or carbonaceous components (such as certain classes of asteroid regolith) may exhibit limited solubility in aqueous systems. For such materials, CSP feasibility becomes conditional, potentially requiring alternative solvent chemistries (e.g., alkaline, chelating, or low-volatility non-aqueous systems) to activate interfacial mass transport.

3.1.2 Fine particulate fraction and surface defect density

Pressure-enhanced mass transport during CSP is strongly facilitated by high surface area, structural defects, and amorphous or glassy phases, which promote dissolution and precipitation at particle contacts (Galotta and Sglavo 2021).

Many extraterrestrial regoliths naturally exhibit fine particle size distributions as a result of micrometeoroid impacts and space weathering processes, generating fractured surfaces and chemically activated sites. From a CSP perspective, such features can be advantageous, provided that suitable solvent–solid interactions are available. This suggests that particle-scale characteristics of extraterrestrial materials may, in some cases, partially compensate for chemical heterogeneity, enhancing CSP responsiveness even in non-basaltic systems.

3.1.3 Pressure-assisted mass transport independent of gravitational loading

Unlike conventional thermal sintering, CSP relies on externally applied mechanical pressure, rather than gravitational settling, to activate densification mechanisms. Experimental studies have demonstrated that CSP densification is dominated by localized stress concentrations, pressure-solution creep, and liquid-mediated transport at particle contacts. As a result, the fundamental densification mechanisms of CSP are not intrinsically gravity-dependent, making the process conceptually compatible with reduced-gravity or microgravity environments. The primary challenge in space applications therefore lies not in the sintering mechanism itself, but in the engineering of pressure delivery, reaction-force management, and load uniformity.

3.1.4 Thermodynamic stability of bonded or precipitated phases at low temperature

For CSP-fabricated materials to be viable in extraterrestrial environments, the newly formed interparticle bonds or precipitated phases must exhibit thermodynamic and environmental stability under vacuum, radiation exposure, and thermal cycling. Previous CSP studies have shown that low-temperature precipitated phases can achieve sufficient mechanical integrity, provided that volatile hydrates or metastable phases are minimized or transformed through controlled post-processing (Rahimi Mamaghani and Parvin 2025a; Li et al. 2025b). This consideration is particularly relevant for space environments, where dehydration, outgassing, or phase decomposition may occur. Consequently, CSP systems intended for extraterrestrial deployment must prioritize solvent selection and reaction pathways that yield stable, low-volatility bonding phases.

3.1.5 Mechanistic classification of CSP applicability

Based on the above criteria, CSP applicability across extraterrestrial materials can be categorized into three broad classes: (i) *Experimentally validated systems*, such as lunar and Martian basaltic regolith simulants consolidated via alkali-assisted CSP; (ii) *Conditionally feasible systems*, including carbonaceous or phyllosilicate-rich asteroid materials, where CSP may be viable subject to solvent compatibility and mineralogical composition; (iii) *Challenging systems*, such as metal-rich asteroid regoliths, where limited solubility necessitates alternative or hybrid liquid-mediated consolidation strategies (Xue and Qiao 2025a). This mechanistic framework for assessing CSP applicability across extraterrestrial material classes is consolidated into a comparative summary in Table 2.

Accordingly, references to asteroid or deep-space CSP applications in this review are intended to provide a mechanistic feasibility framework, rather than to imply direct experimental validation at present. Further experimental investigations using representative simulants and tailored solvent systems will be essential to advance these conditionally feasible scenarios toward higher technological readiness levels.

3.1.6 Influence of regolith weathering state and simulant variability on CSP response

The response of regolith materials to CSP is expected to vary with particle-scale characteristics associated with different weathering layers, including particle fineness, amorphous content, and surface reactivity. Lunar and Martian regoliths are known to exhibit vertical and lateral heterogeneity arising from micrometeoroid impacts, space weathering, and

thermal cycling, resulting in variable proportions of glassy phases, nanophase iron, and fractured mineral surfaces. Studies on lunar regolith simulants have demonstrated that particle fineness and amorphous content significantly influence dissolution kinetics and interfacial reactivity (Fackrell et al. 2024). For example, investigations on JSC-1A simulant have shown that finer fractions exhibit enhanced chemical reactivity and greater participation in low-temperature binding reactions, whereas coarser fractions behave predominantly as inert fillers (Neves et al. 2020). This finding is consistent with CSP theory, in which pressure-assisted dissolution–precipitation is amplified by increased surface area and defect density. Similarly, regolith-based polymer composite studies have demonstrated that mechanical performance is strongly dependent on particle size distribution and surface interactions, even in non-sintered systems (Bao et al. 2023), underscoring the sensitivity of regolith processing routes to weathering-related material variability.

In contrast, AM of regolith followed by high-temperature sintering requires extensive compositional and rheological optimization to accommodate different simulant chemistries and weathering states, as evidenced by recent DLP-based fabrication studies of lunar highland simulants (Aiyeki et al. 2025; Xu et al. 2026). From a CSP perspective, these findings suggest that while absolute densification efficiency and mechanical performance may vary across regolith weathering layers, the underlying CSP mechanisms remain transferable, provided that sufficient fine fractions and chemically responsive phases are present. Consequently, CSP applicability across extraterrestrial regoliths should be evaluated on a material-specific, mechanism-based basis, rather than assuming uniform processing behavior across all simulants or weathering states.

3.2 CSP-additive manufacturing integration: current status and feasible pathways

The integration of the CSP with AM has been proposed as a promising route for fabricating complex geometries at reduced processing temperatures and is recognized as a key emerging opportunity for CSP (Rahimi Mamaghani and Parvin 2025a). However, it is important to clarify that CSP-AM integration is currently at an early experimental and conceptual stage, with demonstrations largely limited to terrestrial laboratory environments. To date, CSP has not been implemented as a fully in-situ consolidation mechanism within space-based AM platforms, and its application in extraterrestrial contexts should therefore be regarded as translational rather than established. This aligns with the broader context of AM for space, which is an emerging but still maturing technology that requires careful development to ensure reliability (Georgantzinis et al. 2025). One feasible and experimentally supported integration pathway

Table 2 Mechanistic assessment of CSP applicability to extraterrestrial material classes

Material class	Dominant phases	Solubility pathway	CSP feasibility	Key Refs.
Lunar basaltic regolith	Oxides, glass	Alkali dissolution	Experimentally validated	Xue and Qiao (2025a)
Martian basaltic Rivera et al. (2023) soil	Oxides, clays	Alkali activation	Experimentally validated	Karacasulu et al. (2023)
Carbonaceous asteroid fines	Phyllosilicates, carbon	Aqueous/non-aqueous	Conditionally feasible	De Gregorio et al. (2024)
Metal-rich asteroid regolith	Fe–Ni alloys	Limited solubility	Challenging; alternative solvents required	Rivera et al. (2023)

involves AM-assisted green body fabrication followed by post-printing CSP consolidation. In this approach, AM techniques such as extrusion-based printing or binder jetting are used to produce porous or weakly bonded structures with controlled geometry (Gupta et al. 2025). These printed bodies are subsequently subjected to CSP under moderate pressure and low temperature, enabling densification and interparticle bonding without exposing the structure to the high thermal loads typical of conventional sintering. This strategy preserves geometric complexity while leveraging CSP for mechanical strengthening and microstructural refinement. Such post-printing consolidation concepts have been demonstrated for ceramic systems under terrestrial conditions and represent a technically realistic pathway for future space adaptation.

A second, less mature integration route involves CSP-assisted layer-wise or localized consolidation during AM, particularly for paste- or slurry-based extrusion processes. In this scenario, mechanical pressure is applied locally, either during deposition or immediately after layer placement, to activate solvent-mediated bonding between adjacent layers. While this approach introduces additional challenges related to pressure delivery and load uniformity, it offers the potential to enhance interlayer adhesion and reduce porosity without global high-temperature processing. At present, this pathway remains at a conceptual or proof-of-principle level, requiring further experimental validation before space deployment can be considered. Overall, CSP-AM integration should be viewed as a complementary hybrid strategy, where AM provides geometric freedom and CSP enables low-temperature consolidation. Its future applicability in space manufacturing will depend on advances in localized pressure application, solvent management, and system integration under reduced-gravity and vacuum conditions. The proven ability of CSP to fabricate advanced polymer-ceramic composites suggests strong potential for creating multifunctional, AM-built structures (Guo et al. 2018).

3.3 Lunar applications

The Moon's surface offers abundant basaltic regolith (a mixture of silicates, oxides, and glass) that can serve as CSP feedstock (Cool et al. 2023). Lunar regolith simulant (like JSC-1A and LMS-1) has been sintered in the lab to produce durable moon bricks (Iantaffi et al. 2025; Gupta et al. 2024; Hossain and Bullard 2025). Key considerations for the lunar environment include vacuum, thermal cycling, and partial gravity (≈ 0.16 g). Practically, leveraging CSP on the Moon will depend on localized solvent availability and containment strategies: polar water–ice resources could supply the transient liquid for CSP in some sites, but processing must occur in sealed dies or low-vapor environments to prevent solvent loss in vacuum and to enable effective closed-loop solvent recovery. One recent study by Liu et al. in 2024 demonstrated energy-efficient CSP of a lunar regolith simulant (Liu et al. 2024). In their experiments, a powdered simulant mixed with a NaOH solution was compressed at 200 MPa and heated to only 150 °C for 90 min. The cold-sintered block achieved a density of 2.44 g/cm³ and compressive strength of 207 MPa values comparable to or exceeding high-temperature sintered regolith bricks. Importantly, this performance held up even after repeated thermal cycling from -196 °C to $+150$ °C, suggesting good reliability under lunar day-night swings. Such strong, dense bricks could be used as habitat walls, radiation shields, or landing pads. For reference, many proposed bagged regolith systems or regolith-based geopolymer construction concepts achieve only a few MPa in compression, primarily because they rely on loose granular confinement or ambient-pressure alkali activation rather than true densification. Recent studies on bagged regolith shielding systems and cast geopolymer binders report compressive strengths typically below 5–10 MPa, which are sufficient for radiation shielding or berm stabilization but inadequate for load-bearing structural components (Wu et al. 2024; Bao et al. 2024). In contrast, the cold-sintered regolith blocks discussed above exhibit compressive strengths exceeding 200 MPa, representing an improvement of nearly two orders of magnitude. This comparison highlights that CSP enables a

fundamentally different class of lunar construction materials, transitioning from passive shielding solutions to mechanically robust, load-bearing infrastructure. The Moon has the advantage of abundant solar power and (at some sites) water ice (Zhang et al. 2024b). Water or hydroxide, as a common solvent in CSP, can likely be sourced from polar ice or hydrated minerals (Sun 2022; Song et al. 2021). The vacuum complicates CSP, since volatile solvents will evaporate quickly. Most recently, Xue et al. (2025a) introduced CSP to produce high-performance lunar regolith simulant-based geopolymers as a low-energy alternative for lunar construction. They found that by optimizing alkali content and water addition, CSP enables rapid densification via pressure-solution creep, raising compressive strength from 54.56 MPa to 205.87 MPa within 60 min while reducing water consumption by up to 90% compared to conventional casting. The results demonstrate CSP's potential as an energy-efficient, resource-saving strategy for sustainable in-situ lunar infrastructure development. In practice, CSP on the Moon would require a sealed press or pressure vessel, or possibly using low-vapor-pressure solvents (Azami et al. 2024). In the lab experiment, processing was done in air, but future lunar CSP units might use infrared lamps or electric heaters inside a glovebox to control the environment. The low lunar gravity means that natural compaction is weaker than on Earth; however, CSP applies explicit pressure to densify the bricks. Indeed, sintering in microgravity tends to be less effective due to the lack of hydrostatic stress, so mechanical pressing is even more crucial on the Moon.

From a mechanistic perspective, the effectiveness of NaOH- and alkali-based solvents in CSP is closely related to dissolution-precipitation reactions and alkali activation chemistry, which have been extensively studied in construction materials and geopolymer systems. Alkali solutions promote rapid dissolution of aluminosilicate and silicate phases, followed by reprecipitation and interparticle bonding under confined conditions. Recent studies have shown that even at relatively low temperatures, concentrated alkali environments significantly enhance mass transport and particle rearrangement, leading to accelerated densification and strength development when combined with external pressure (Geng et al. 2024; Xiong et al. 2022). In particular, pressure-assisted alkali activation enables substantial reduction in water content while maintaining or even improving mechanical performance, a feature that is critical for extraterrestrial applications where solvent availability is severely constrained (Geng et al. 2024). These findings support the interpretation that, in lunar regolith CSP systems, alkali solvents such as NaOH act not only as transient liquids but also as chemically active agents that facilitate pressure-solution creep and solid-state bonding at low thermal budgets.

Beyond simple bricks, CSP could fabricate functional components on the Moon. For example, mixtures of regolith

with conductive additives could yield sensor housings or structural electrodes, and regolith-based glass or ceramics might be transparent or reflective for optical uses (Ellery 2025). Also, CSP can integrate fibers or mesh (perhaps made from metal or polymer brought from Earth) into regolith composites, creating reinforced structures that can withstand micrometeoroid impacts. The multi-material aspect of CSP allows a top-down approach to building walls with graded properties. For instance, an inner layer optimized for structural strength and an outer layer enriched in high-Z elements for radiation shielding. By adjusting the solvent chemistry, CSP could even activate in-situ minerals (like using native alkalis in the regolith) to function as binders. In summary, on the Moon, CSP provides a low-energy path to densify native soil into strong building materials. Experimentally cold-sintered lunar bricks have achieved very high strength and stability under lunar-like thermal conditions. These results suggest that CSP could form the basis of future lunar construction, producing everything from habitat blocks to runway pavements, with far less power than oven-sintering.

3.4 Martian applications

Mars presents a different set of resources and constraints. The Martian surface has basaltic, clay-rich regolith with perchlorate salts and an atmosphere of CO₂ at a low average surface pressure of approximately 610 Pa (Ray et al. 2025; Corchete 2024). Water is present, either as ice or bound in minerals (Nazari-Sharabian et al. 2020). The CSP on Mars has been studied using Earth-made Martian regolith simulants. Karacasulu et al. (2023) report the first successful CSP of Mars simulant MGS-1. They mixed the simulant (< 100 µm particles) with NaOH solution and applied pressures up to 250 MPa at temperatures of 150–250 °C for 30 min; higher NaOH concentrations and temperatures produced denser compacts. Finally, at 250 °C with 10 M NaOH, they achieved ~ 88% theoretical density and compressive strength ~ 45 MPa. These values, though lower than the lunar CSP case, still represent useful building materials from Martian soil. Even an 88% density brick could serve as a block or base layer for a structure. The achievable strength (~ 45 MPa) is on par with some concrete mixtures and far above loose soil or simple cements. However, it indicates that Martian soil chemistry can be exploited to produce structural components with modest processing windows. Mars-specific advantages include accessible water and chemical reactions; perchlorates or other salts in Mars soil can be used to produce alkaline solutions (NaOH, KOH) in situ by electrolysis or reaction, so the needed solvent/binder could potentially be made from local materials (Pajares et al. 2025). Also, the Martian atmosphere and pressure mean that processing could occur at partial pressure rather than full vacuum. However,

Mars has dust storms and temperature swings ($-100\text{ }^{\circ}\text{C}$ to $20\text{ }^{\circ}\text{C}$), so any CSP device must cope with thermal insulation and dust control. The gravity on Mars ($\sim 0.38\text{ g}$) still provides some particle compaction, but a press would again be required for densification.

In parallel with pressure-based densification routes, AM has been widely investigated for Martian construction due to its flexibility, reduced formwork requirements, and compatibility with in-situ resource utilization (Janabadi et al. 2025). Multiple studies have demonstrated the feasibility of 3D printing structural elements using Martian or lunar regolith simulants, employing extrusion-based deposition, binder-assisted printing, or hybrid thermal consolidation approaches (Mannes Guesser De Oliveira et al. 2025; Korniejenko et al. 2022). Recent work has emphasized the challenges of off-world AM, including powder handling under reduced gravity, binder transport and curing, and thermal management in low-pressure environments (Geng et al. 2025). Polymer-assisted and composite-based printing strategies have also been proposed to enhance green strength and shape fidelity, particularly for complex geometries and modular construction elements (Li et al. 2020). While promising, these approaches often require imported binders, significant thermal input, or post-processing steps to achieve structural-grade properties. Beyond AM, high-temperature in-situ fabrication techniques have also been explored for Martian and lunar environments. One notable approach involves thermite or geothermite reactions, which generate sufficient localized heat to melt and consolidate regolith-derived materials without external furnaces. Faierson et al. demonstrated the fabrication of shaped structural elements by initiating a geothermite reaction in regolith simulant mixed with aluminum powder, producing dense components that retained the geometry of the containment mold (Faierson et al. 2010). Such methods offer rapid consolidation and minimal reliance on external power sources but involve extreme thermal gradients, limited dimensional control, and challenges related to scalability and repeatability. While these approaches demonstrate the feasibility of extraterrestrial manufacturing, they generally require high thermal budgets, imported binders, or complex thermal control systems. In contrast, CSP offers a fundamentally different pathway based on pressure-assisted densification at low temperatures, motivating its consideration as a complementary or alternative strategy for off-world construction.

Potential Martian applications of CSP include constructing habitats and infrastructure from regolith. For example, wall blocks could be printed or cast in modules, then pressed. The CSP bricks might be impermeable enough to hold fuels or water, acting as storage tanks. Because CSP processing temperatures are well below the boiling point of many organic spacecraft fluids, CSP ovens could potentially reclaim waste polymers or plastics by co-sintering them with

regolith (a form of recycling) (Lai et al. 2024a). Another idea is integrating radiation-shielding tiles in the floor or walls by co-sintering iron-rich soil fractions or adding metal powders. In brief, CSP offers Mars explorers a way to convert Martian soil into reliable building blocks. The initial case study shows that even basic Martian simulant can be cold-sintered into decently strong parts. With further optimization of chemistry and processing, CSP could yield a variety of Martian construction materials, complementing geopolymers and sintered slags, but with much lower energy cost.

Most experimental demonstrations of CSP for extraterrestrial construction have relied on lunar and Martian regolith simulants rather than returned samples, making the representativeness of these materials an important consideration (Xue and Qiao 2025a). Widely used simulants are generally designed to reproduce the bulk oxide composition, dominant mineral phases, and particle size ranges of their extraterrestrial counterparts, which are key parameters governing mechanical response and pressure-assisted densification (Slabic et al. 2024). As a result, simulants are effective for evaluating CSP feasibility in terms of achievable density, strength, and processing windows under controlled laboratory conditions. However, important differences remain between simulants and real regolith that may influence CSP behavior. In particular, simulants typically lack the extreme surface disorder and nanophase features induced by space weathering, such as nanophase metallic iron, amorphized grain rims, and radiation-induced defects, which can modify surface energy, wettability, and dissolution kinetics (Xiong et al. 2024). Additionally, the glassy fraction and chemical heterogeneity present in natural regolith may be spatially more complex than in engineered simulants, potentially affecting solvent-mediated mass transport and interparticle bonding during CSP. Consequently, while simulant-based studies capture the dominant densification mechanisms of CSP, they may not fully represent the magnitude or variability of reaction kinetics and mechanical performance achievable in situ. Importantly, CSP is governed less by planetary origin than by material-specific criteria such as partial solubility, fine particle size, and the presence of reactive or amorphous phases (Yamaguchi et al. 2025). When these criteria are satisfied, the fundamental CSP mechanisms remain transferable across simulants and real regolith, even if quantitative outcomes differ. The use of advanced simulants incorporating space-weathering analogues, as well as future validation using returned extraterrestrial samples, will be essential for refining CSP process windows and improving predictive accuracy for off-Earth deployment.

3.5 Orbital (microgravity) environments

In Low Earth Orbit or cislunar space (e.g., on the ISS, Lunar Gateway, or free-flying stations), one could imagine bringing small regolith samples (from missions) or using recycled materials to fabricate components via CSP (Chen and Li 2025). The absence of gravity changes the sintering dynamics; as noted above, microgravity can lead to less natural densification in conventional sintering because there is no self-weight to close pores (D'Angelo et al. 2021; Torresani et al. 2022). For CSP, this means the external press must do all the work of compacting particles. Practically, an orbital CSP machine would be a heavy piece of equipment, likely a hydraulic or electromechanical press with thermal control, installed in a compartment to contain any solvents or dust. Because space habitats maintain a few hundred mbar or 1 bar internal pressure, CSP could be done without vacuum complications (unlike on the Moon), but care must be taken to avoid venting toxic solvents (NaOH vapor is dangerous). The ambient temperature may need to be raised (such as via resistive heaters) to reach the needed 100–200 °C, but heat rejection is less of a concern with spacecraft radiators.

Applications in orbit might include manufacturing spare parts and tools from ceramic materials. For example, cold-sintered insulation tiles, ceramic bearings, or protective panels using powdered waste or imported regolith. The CSP could also be used to create radiation shielding domes by forming thick ceramic shells on modules. The microgravity environment allows building large hollow or lightweight lattice structures that would collapse under Earth's gravity; CSP could solidify or reinforce these structures by binding metal or polymer frameworks with ceramic. Additionally, orbital CSP systems could support on-demand fabrication; astronauts could mix powder and binder in a mold, then press it into shape, much like a 3D printer with a post-sinter cycle (Liu et al. 2025). One speculative use is using CSP to fabricate optical or electronic components. Because CSP can integrate with many materials, an orbital lab might cold-sinter a piezoelectric wafer or a superconducting oxide under controlled conditions, enabling electronics experiments beyond Earth. Another is space debris recycling; ground-up micrometeoroid shielding or alloy powders could be bound into new shapes via CSP, closing a materials loop. While no specific orbital CSP demonstration exists yet, the concept benefits from established space manufacturing programs. For instance, the ISS has flown furnace units to sinter metals and ceramics (like the Low Gradient Furnace) (Karl et al. 2020). We therefore recommend a focused technology demonstration: a compact CSP press and sealed processing chamber operated aboard the ISS or Gateway to validate pressure-assisted densification, solvent recovery, and sensor-based closed-loop control under reduced gravity and controlled atmosphere before any surface deployment. A

CSP device could be a logical extension, enabling pressured low-temperature sintering in zero-g. Given the limited payload capacity to orbit, such a device would likely focus on high-value or necessary parts (like medical ceramics or specialized tiles) rather than large building blocks. Nonetheless, the general principle of performing CSP in microgravity is sound, provided one accounts for the lack of buoyant flow and ensures a uniform applied pressure.

3.5.1 Space adaptability of CSP equipment in orbital and microgravity environments

The deployment of CSP in orbital or deep-space environments introduces engineering constraints distinct from those encountered on planetary surfaces. In microgravity or near-zero-gravity conditions, gravitational compaction is negligible, and all densification forces must be externally imposed and mechanically reacted within the system. Importantly, the fundamental mechanisms governing CSP (pressure-assisted dissolution–precipitation, interfacial mass transport, and stress-induced particle rearrangement) are not intrinsically gravity-dependent (McPherson and DeLucas 2015). Instead, densification is driven by localized contact stresses generated through applied mechanical pressure. This distinguishes CSP from gravity-assisted powder processing routes and suggests conceptual compatibility with microgravity environments. The primary engineering challenge for orbital CSP implementation lies in the generation and management of compressive loads. In the absence of gravitational reaction forces, CSP systems must employ closed-loop or self-reacting mechanical architectures, such as opposed-anvil presses, frame-supported die assemblies, or internally braced reaction structures, to ensure load balance and prevent momentum transfer to the hosting spacecraft or station (Meléndez et al. 2025). Similar force-management strategies are already employed in space-qualified mechanical systems, including docking mechanisms and material testing rigs.

Vacuum compatibility further imposes constraints on solvent handling and thermal management. CSP systems intended for orbital deployment must utilize low-volatility solvents, sealed processing chambers, or controlled micro-atmospheres to prevent solvent loss and contamination (Xue and Qiao 2025b). Thermal regulation, although less demanding than in high-temperature sintering, remains essential to ensure consistent liquid-mediated transport and to mitigate localized heating due to frictional or reaction-related effects. Compared with alternative space manufacturing technologies such as laser or solar sintering, CSP offers potential advantages in orbital contexts by avoiding extreme temperatures, high optical power densities, and open-beam energy delivery. However, CSP implementation in orbit remains contingent on advances in compact press design,

solvent recycling strategies, and autonomous process control. Accordingly, CSP should be regarded as mechanistically compatible but technologically emerging for orbital manufacturing, with further system-level development required to reach higher readiness levels.

3.6 Deep-space and asteroid applications

In deep space (beyond Earth's magnetosphere) or on small bodies (asteroids, moons of Mars, etc.), CSP could play a key role in both construction and ISRU missions. Many asteroids contain regolith-like material and significant water or ice, particularly carbonaceous asteroids. If a future mission aimed to extract ore or build a station on an asteroid, CSP could consolidate asteroidal powder into safe or useful structures. The extremely low gravity on asteroids (milligravity) makes conventional building impossible, but CSP does not rely on gravity at all; it only needs local pressure. One could anchor a press to the surface and compress mined rock powder into a boulder-shaped habitat module. If water ice is present, melting a fraction to serve as the solvent could eliminate the need to carry chemical binders. Moreover, asteroids often have high carbon or metal content; CSP formulations might be adjusted (using ethanol, ammonia, or non-aqueous deep eutectic solvents) to handle non-oxide materials, potentially enabling in-space sintering of metal powders to create bulk components like reactor shields or propellant tanks. Deep-space stations (e.g., a crewed vehicle traveling to Mars or a Phobos base) could similarly use CSP to fabricate parts from cargo-returned lunar or asteroidal soil. For instance, a cargo capsule arriving at a Mars station could carry a payload of processed regolith pellets and a small CSP press to build a spare part on demand. Furthermore, CSP can integrate with isotopic or chemical processes, cold-sintered nuclear-fuel matrices mixed with lunar uranium-bearing minerals, forming reactor fuel elements without high-temperature meltdown.

For carbonaceous or metal-rich asteroid regoliths, non-aqueous transient phases (for example, alcohols or deep-eutectic solvents (Rivera et al. 2023)) and low-volatility ionic liquids are promising alternatives to aqueous systems and may broaden CSP's applicability beyond oxide-rich lunar and Martian feedstocks. Multidisciplinarily, CSP in these environments invites space architecture planning; structures might be designed to take advantage of CSP-produced shapes and densities. For instance, a station wall could be cold-sintered in layers to embed conduits or fiber optics within the ceramic, creating a multifunctional shell. Radiation shielding is particularly critical in deep space; thick CSP regolith walls (> 1 m) could stop solar and galactic particles effectively. Recent work in radiation shielding shows that dense materials like aluminum or concrete must be tens of centimeters thick to be effective on Mars or Moon (Al Zaman and

Kunja 2023; Meurisse et al. 2020); CSP-molded regolith, being similar in density, would follow the same design rules. Additionally, the mineral composition of regolith (with high iron and silicon) naturally provides good X-ray and particle attenuation. Overall, CSP's applicability spans the solar system; on airless bodies, it offers a closed-system method to consolidate local soil, in microgravity, it allows additive/consolidation manufacturing of critical parts, and in deep space, it can recycle materials into protective and functional architecture. Each environment imposes unique constraints (solvent choice, thermal management, pressure application), but the core CSP technique (using pressure and a solvent to bond powder) remains flexible enough to adapt. A comparison of processing energy, temperature, and compressive strength between conventional and CSP of regolith is shown in Fig. 3. A detailed comparison of experimental results obtained for lunar and Martian simulants using CSP and traditional sintering is provided in Table 3. Although CSP densification mechanisms (namely pressure-assisted dissolution–precipitation and particle rearrangement) remain fundamentally consistent across regolith simulants, the quantitative outcomes of CSP are influenced by regolith maturity, glass content, and secondary mineral phases (Karacasulu et al. 2023). Studies comparing immature basaltic simulants with more mature or glass-rich analogs indicate that higher amorphous content and surface defect density can facilitate solvent-mediated mass transport, while sulfate- or clay-rich components may alter dissolution kinetics and densification efficiency (Cool et al. 2023; Zheng et al. 2024). Importantly, these variations affect the magnitude of achievable density and mechanical strength rather than the feasibility of CSP itself, suggesting that CSP exhibits mechanistic robustness but composition-dependent performance. Similar conclusions have been reported for regolith-based construction materials produced via geopolymerization and alkali activation, supporting the need for simulant-diverse assessments in ISRU-related processing studies (Pederson et al. 2025).

3.7 Guidelines for solvent selection in CSP under extraterrestrial conditions

The CSP fundamentally relies on a transient solvent phase to enable pressure-assisted dissolution, mass transport, and interfacial reprecipitation at temperatures far below those required for conventional sintering (Guo et al. 2016a). In extraterrestrial environments, solvent selection becomes a central design parameter because it simultaneously governs densification kinetics, phase evolution, process stability, and system-level feasibility under constraints such as reduced pressure, extreme thermal cycling, and limited material availability. Consequently, solvent choice for CSP in space must be guided by mechanistic compatibility with the processed

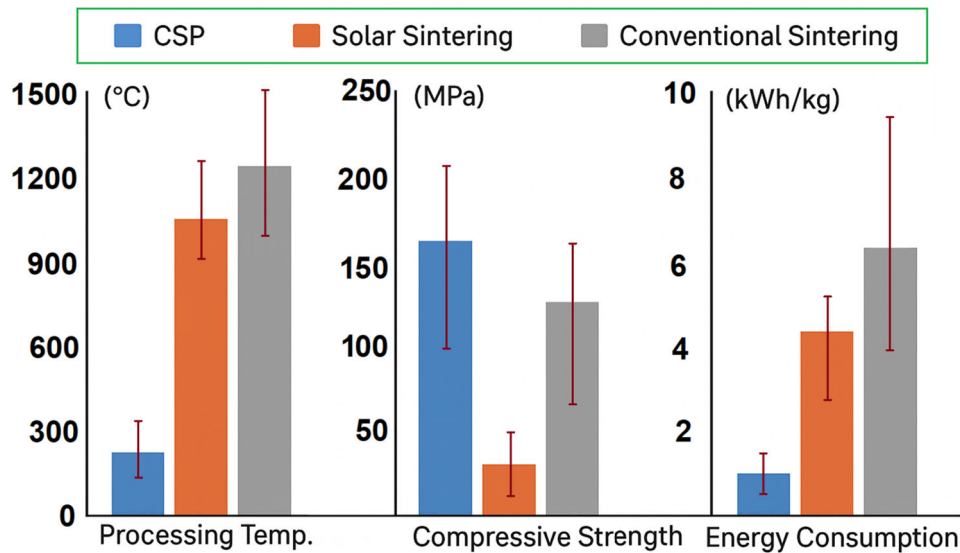


Fig. 3 Comparative performance of regolith processing methods based on representative literature data. Error bars indicate the approximate range of values reported across experimental studies for conventional high-temperature sintering (Taylor et al. 2016; Warren et al. 2022; Ibn-Mohammed et al. 2019), solar sintering (Meurisse et al. 2018; Lim et al. 2021; Rosa 2019), and CSP (Liu et al. 2024; Guo et al. 2016a;

Karacasulu et al. 2023; Xue and Qiao 2025a), reflecting variability in simulant composition and processing conditions rather than statistical uncertainty from a single dataset. The CSP demonstrates substantially reduced processing temperature and energy consumption while achieving competitive or superior compressive strength compared with solar sintering and conventional high-temperature sintering

material as well as by environmental boundary conditions, rather than by terrestrial processing conventions alone.

From a mechanistic perspective, suitable CSP solvents must satisfy several core criteria. First, the solvent should exhibit partial chemical compatibility with the solid phases, enabling limited dissolution of reactive constituents such as silicate glass, alkali-bearing phases, or surface-hydroxylated species, while avoiding excessive leaching or uncontrolled chemical degradation. This dissolution–precipitation cycle is driven by pressure–solution creep (Xue and Qiao 2025a). Second, solvent volatility and vapor pressure must be carefully considered, as high-vapor-pressure solvents may rapidly evaporate under vacuum or low-pressure conditions unless enclosed within sealed or pressure-assisted processing chambers. In open-system CSP, solvent evaporation can significantly increase supersaturation and affect microstructural evolution (Ndayishimiye et al. 2020b). Third, thermal stability and freezing behavior are critical, particularly in environments subject to extreme temperature fluctuations, such as the lunar surface; solvents must remain chemically active within the CSP temperature window or be paired with modest thermal conditioning. Finally, the solvent must respond favorably to applied pressure by enhancing interfacial transport and precipitation kinetics, rather than suppressing densification or promoting brittle secondary phase formation. From an ISRU perspective, solvent availability, recoverability, and recyclability further constrain viable choices.

Material composition strongly influences optimal solvent selection. For silicate- and glass-rich lunar regolith, which is largely anhydrous and dominated by impact-generated amorphous phases, water-based solvents or dilute alkali solutions are generally effective, as they promote controlled surface dissolution and interparticle bonding without triggering extensive chemical reactions. Studies demonstrate that optimizing water (3–11 wt%) and alkali content in the sintering liquid can significantly enhance densification and compressive strength (Xue and Qiao 2025a). In contrast, Martian regolith contains higher fractions of hydrated minerals, sulfates, and chemically reactive fines, which can significantly modify solution chemistry during CSP. In such systems, alkali-modified aqueous solvents with carefully controlled concentration are advantageous, as they buffer pH variations and stabilize dissolution–precipitation behavior while minimizing excessive geopolymerization or salt crystallization. Proof-of-concept studies have successfully densified Martian regolith simulants using NaOH solutions as the liquid medium (Karacasulu et al. 2023). Asteroid-derived materials present the greatest variability, ranging from anhydrous silicates to volatile- and organic-rich carbonaceous assemblages. In these cases, CSP feasibility is inherently conditional, and solvent selection may require hybrid or low-activity systems, such as diluted aqueous solutions or low-volatility mixed solvents, tailored to the specific mineralogical composition.

Table 3 Comparative results from CSP and conventional regolith processing methods using lunar and Martian simulants

Environment	Feedstock	Processing conditions	Density	Compressive strength	Energy consumption	Key notes on consistency/WEATHERING EFFECTS	Refs.
Lunar (simulant)	LMS-1 (basaltic, immature lunar simulant) with NaOH solution	CSP: 200 MPa, 150 °C, 90 min	2.44 g/cm ³	207 MPa	0.86 kWh	High densification attributed to fine particle size and reactive basaltic phases; CSP mechanism robust for immature simulants	Liu et al. (2024)
Lunar (simulant)	HIT-L-2 (basaltic, compositionally distinct lunar simulant) with NaOH solution	CSP: 200 MPa, 200 °C, 60 min	2.54 g/cm ³	206 MPa	~ 70–90% lower water use compared to conventional routes	Comparable mechanical performance despite simulant differences, indicating mechanistic consistency of CSP	Xue and Qiao (2025a)
Martian (MGS-1)	MGS-1 (basaltic Martian simulant with altered mineral phases) + NaOH	CSP: 250 °C, 30 MPa, 30 min	~ 88% theoretical density	~ 45 MPa	N/A	Reduced pressure reflects compositional differences; CSP feasibility retained but performance is composition-dependent	Karacasulu et al. (2023)
Lunar (solar)	JSC-1 (basaltic lunar simulant)	Solar sintering: Focused solar, 1000–1200 °C	Porous	< 5 MPa	High (> 5 kWh)	High-temperature processing yields poor mechanical integrity; strongly energy- and environment-dependent	Meurisse et al. (2018)

Environmental context further refines solvent choice. On the lunar surface, near-vacuum conditions significantly increase evaporation rates, necessitating sealed CSP chambers and solvent recovery strategies, even when water-based systems are used (Xue and Qiao 2025a). On Mars, the low-pressure CO₂ atmosphere reduces convective heat transfer and enhances evaporation, but still permits the use of aqueous or alkali-modified solvents when moderate pressure and thermal control are applied. In orbital and microgravity environments, solvent behavior is dominated by surface tension and capillary forces rather than gravity, requiring fully enclosed, closed-loop systems to prevent free-fluid migration. In such settings, low-volatility solvents and precise solvent dosing become particularly important for maintaining process stability and uniform densification. Taken together, these considerations demonstrate that CSP solvent selection for extraterrestrial applications is not governed by a universal solution, but by a mechanistically informed matching of solvent chemistry, material composition, and environmental constraints. When such alignment is achieved, CSP offers a uniquely adaptable and energy-efficient pathway for in-space materials fabrication, whereas inappropriate solvent choice can undermine densification efficiency, mechanical performance, and system reliability.

4 Challenges

This section focuses on the primary material, process, and environmental challenges associated with implementing CSP in space, while engineering solutions and technology-enabled mitigation strategies are discussed in Sect. 5. Figure 4 schematically illustrates the key environmental constraints (such as gravity, atmosphere, and solvent accessibility) that uniquely affect CSP implementation across different extraterrestrial settings. While promising, CSP in space faces several significant challenges. At present, CSP has not been experimentally validated for non-oxide-rich asteroid materials, and extending CSP to metal-rich or carbonaceous regoliths will require tailored solvent systems and further experimental verification (see Table 2). These span materials, process engineering, and environmental factors:

4.1 Solvent and binder availability

The CSP typically uses water or aqueous solutions (e.g., NaOH) as the transient liquid. On the Moon, liquid water is scarce except in shadowed craters. Mars has more water, but regolith also contains harmful perchlorates (Rzymiski et al. 2024). Space stations have water but must conserve it. Solutions include importing binders (mass penalty) or generating them in situ (e.g., electrolyzing ice to make NaOH). Alternatively, low-vapor liquids like alcohols or ionic liquids could

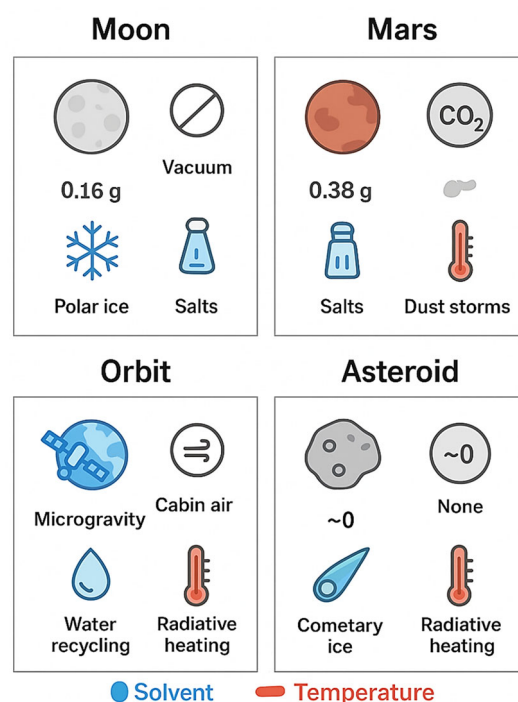


Fig. 4 Environmental factors influencing CSP performance across lunar, Martian, and asteroid environments (gravity, atmosphere, temperature, solvent availability)

be explored, though they may react differently with oxides. In any case, a closed-loop recycling of solvent will be needed to avoid loss into the vacuum or repeated resupply. Similar closed-loop solvent management concepts and low-volatility working fluids have been extensively investigated in space life-support systems and space chemical processing, providing relevant design precedents (Karacasulu et al. 2023; Delorme et al. 2025; Shiwei et al. 2020).

4.2 High pressure requirements

Most CSP demonstrations use 100–300 MPa of pressure to achieve good density (Evangeline and Annamalai 2025; Mamaghani and Parvin 2023). Generating such pressures in situ demands heavy, robust presses or dies. Bringing large hydraulic presses to the Moon or Mars is challenging. Besides, recent developments in energy storage systems for space (Sharma and Santasalo-Aarnio 2025), potential solutions include using stored mechanical energy (spring or flywheel), multiple smaller presses, or even explosive compaction (if safety allows). Pneumatic or chemical high-pressure gas could also be considered, but any system adds complexity and weight. Innovation is needed to design lightweight, energy-efficient pressing hardware for space. In addition to hydraulic systems, novel compact actuation concepts, such as pre-loaded springs with incremental

release, mechanical advantage gantries, or modular roll-compaction staging, should be explored to reduce mass and power requirements of space-flown pressing systems. Potential solutions include using stored mechanical energy (such as spring or flywheel-based systems), concepts that have been proposed or demonstrated in compact mechanical systems for space and terrestrial high-force applications (Karacasulu et al. 2023; Lindmark et al. 2509). These pressure-generation challenges directly motivate the need for lightweight, internally balanced press architectures, which are discussed in the following subsection.

4.3 Engineering pathways for lightweight CSP systems in space

While the concept of lightweight CSP equipment is often discussed at a high level, its practical realization in space environments requires a combination of structural, mechanical, and system-level design strategies. Importantly, “lightweighting” in space does not necessarily imply reduced load capacity, but rather optimized mass-to-force efficiency and internal force balancing. Structural lightweighting strategies may include the use of high-specific-strength materials (e.g., aluminum–lithium alloys, titanium alloys, or fiber-reinforced polymer composites) for press frames and load-bearing components, as well as topology-optimized structures that minimize mass while preserving stiffness under compressive loading. For instance, modern aluminum–lithium alloys can offer approximately 40–45% weight savings compared to conventional aerospace aluminum, with each 1% lithium addition reducing density by ~ 3% and increasing elastic modulus by ~ 6% (Hajjioui et al. 2023). Additively manufactured metallic components with lattice or truss architectures can further reduce mass while maintaining mechanical integrity. Self-reacting and closed-frame press architectures represent a critical pathway for CSP deployment under microgravity conditions. In such systems, compressive forces are internally balanced between opposing platens or anvils through a rigid or preloaded frame, eliminating reliance on external reaction forces or gravitational loading. Similar principles are employed in compact mechanical testing rigs and docking mechanisms designed for space applications. This approach is essential because the densification behavior of granular materials, a process relevant to powder-based CSP, is fundamentally altered in microgravity, with jamming transitions occurring at lower packing fractions compared to terrestrial conditions (D’Angelo et al. 2022).

4.4 Uniform-pressure CSP systems in space

Pressure uniformity control, which is essential for consistent CSP densification, can be achieved through multiple

complementary approaches. These include the use of compliant interlayers (e.g., elastomeric or polymeric pads), segmented or floating platens that self-align under load, and quasi-isostatic pressure distribution systems in which force is transferred through deformable media. Such strategies reduce stress concentrations and compensate for minor misalignments, which are particularly critical in microgravity environments where passive settling is absent. The effectiveness of compliant, self-aligning mechanisms for achieving uniform load distribution in space is demonstrated by innovative solar concentrator designs that use elastic hinge mechanisms and custom-shaped ribs to achieve up to 92.24% uniformity in energy flux density, a principle analogous to uniform pressure application (Li et al. 2024). Although these systems address different functional objectives, they demonstrate the feasibility of achieving controlled force or energy uniformity in lightweight, deployable space structures, a principle directly relevant to CSP press design. Actuation mechanisms suitable for space-based CSP presses may include electromechanical screw-driven actuators, piezoelectric stacks for fine pressure control, or hybrid hydraulic systems operating within sealed loops (Ding and Zhou 2025). These approaches allow precise force application and feedback-controlled pressure regulation without introducing excessive system mass or complexity, a principle validated by other space-based systems that prioritize “flexible, lightweight structures” and integrated control circuits to manage forces. Collectively, these engineering solutions suggest that lightweight, uniform-pressure CSP systems are technically feasible in space, provided that press design prioritizes internal force closure, compliant load transfer, and active pressure control. While such systems remain at a conceptual-to-laboratory development stage, they are grounded in established mechanical engineering principles and emerging experimental evidence from granular physics in microgravity and analogous space-deployable structures.

4.5 Particle size and preparation

Similar trade-offs between comminution (grinding) energy and downstream processing efficiency are a fundamental consideration in designing ISRU systems for lunar construction (Vasudeva et al. 2026). The CSP works best with fine powders (< 100 μm) to ensure good packing and dissolution (Rahimi Mamaghani and Parvin 2025a). Regolith as mined is often coarse and irregular. Grinding or milling equipment would be required to prepare the feedstock, adding extra equipment. Alternatively, sieving or attrition in water might generate the needed fines. The energy cost of powdering must be balanced against CSP’s energy savings. Ideally, a synergy could be found, like using grinding waste (which already produces fines) or natural micro-particle fractions.

4.6 Temperature control in vacuum

On the Moon and in free space, thermal control is tricky. To maintain the 100–250 °C required for CSP, an enclosure with heating (and possibly cooling) is needed. The large temperature swings (like lunar night) could help by radiating heat at cycle end, but during processing, the hot block must not lose temperature too quickly (Dong et al. 2022). The device may need insulation and vacuum jackets. Also, in the vacuum the solvent will evaporate strongly unless contained; a sealed die and pressure vessel is necessary. This increases complexity compared to Earth labs. Thermal insulation, vacuum-compatible heaters, and sealed processing enclosures are well-established challenges in space manufacturing and materials testing systems (Vasudeva et al. 2026).

4.7 Gravity and microgravity effects

Building on the engineering considerations discussed above, gravity and microgravity effects introduce additional process-level challenges specific to CSP. As discussed, low gravity tends to reduce natural compaction and can allow pores to grow, due to the buoyancy of vapor and lack of overburden (D'Angelo et al. 2021). While CSP's externally applied pressure solves much of this, the process dynamics can still differ: for instance, liquid can distribute unevenly without gravity, and bubbles or vapor pressures might not escape as easily. Mixing powders and liquids without settling may also be harder in microgravity. Thorough testing (perhaps on parabolic flights or the ISS) will be needed to characterize CSP behavior in these regimes.

4.8 Material heterogeneity and chemistry

Natural regolith is chemically complex (multiple oxides, glass, metal grains) and can contain impurities (such as perchlorates, sulfates, lunar ilmenite) (Ramkissoon et al. 2019). These differences affect solubility and reaction with the binder. For example, chlorides or sulfates could precipitate as salts, changing the pore structure. Each new regolith source may require customizing the CSP recipe (choice of solvent, concentration, additives) to achieve good binding without deleterious reactions. The dissolution step must not simply leach out valuable elements or create unwanted byproducts.

Unlike conventional thermal sintering, the CSP is governed by solution-mediated mass transport, interfacial reactions, and mechanically assisted particle rearrangement (Ndayishimiye et al. 2020a). As a result, regolith impurities can exert a disproportionate influence on CSP behavior through multiple coupled mechanisms. First, soluble ionic impurities, such as sulfates, chlorides, and alkali salts commonly reported in Martian and lunar regolith simulants,

can significantly modify the local solution chemistry during CSP (Edgar et al. 2022). These species may alter pH, ionic strength, and solubility equilibria, thereby accelerating or suppressing dissolution–precipitation pathways that are central to CSP densification. In some cases, impurity-derived ions can act as transient mineralizers, promoting interparticle bonding; in others, they may inhibit neck formation by stabilizing competing aqueous complexes. Second, secondary phase formation induced by impurities plays a critical role. For example, reactions between alkali activators and silicate-rich glassy phases or sulfate-bearing minerals can generate hydrated or amorphous intergranular phases. While such phases may enhance initial densification at low temperatures, they can also introduce mechanically weak interfaces or long-term stability concerns, particularly under cyclic thermal or radiation loading relevant to space environments (Zheng et al. 2024). Third, surface-active impurities, including fine dust fractions, nanophase iron, and amorphous coatings typical of space-weathered regolith, can modify particle surface energy, wettability, and adsorption behavior (Xiong et al. 2024; Zhang et al. 2025). These effects directly influence capillary forces, liquid film stability, and interfacial mobility during CSP, thereby affecting pressure-assisted particle rearrangement and densification kinetics. Finally, mechanical heterogeneity introduced by hard or metallic inclusions (e.g., Fe–Ni particles) can generate localized stress concentrations under applied pressure. In CSP, where densification occurs at relatively low temperatures, such stress localization may lead to non-uniform compaction, microcrack initiation, or anisotropic mechanical properties in the consolidated body (Karbala et al. 2025). Collectively, these mechanisms indicate that impurity effects in CSP are not merely compositional noise, but active contributors to densification pathways, microstructural evolution, and final performance. This sensitivity also suggests that controlled utilization or selective mitigation of impurities, rather than complete purification, may represent a viable strategy for optimizing CSP in ISRU scenarios.

4.9 Cycle and durability

Space structures must survive extreme cycles (thermal, mechanical vibration, radiation). CSP materials need testing under these conditions. The lunar CSP block maintained strength after a few cycles (Liu et al. 2024; Xue and Qiao 2025a), but space radiation or long-term UV exposure could cause surface degradation or changes in composition. If organics or polymers are co-sintered, off-gassing under vacuum and radiation breakdown are concerns. Characterizing long-term creep, fatigue, and fracture behavior under space conditions is crucial. Such degradation mechanisms for polymers and composites are a recognized challenge that must

be accounted for in the design of long-duration lunar surface structures (Zhang et al. 2023).

4.9.1 Scale-up and additive integration

Moving from lab-scale (centimeter-size bricks) to meter-scale structures is nontrivial. Scaling up CSP (Jabr et al. 2023) could be applied to space applications; however, careful control of punch alignment and heating rates would be crucial to achieve defect-free parts in microgravity. The CSP is usually done in static presses; building a whole habitat wall would require either joining many pressed blocks or developing continuous processing (such as a moving belt press, or a heated mold that is printed and compressed layer by layer). Joining blocks (like mortar in bricklaying) adds complexity (Marín-Cortés et al. 2024), potentially requiring separate adhesives or thermal bonding. Integrating CSP with 3D printing (for example, printing a paste that is then pressed and cured) remains at the concept stage. Machines that combine extrusion, layer consolidation, and sintering in one automated operation must be engineered. While such integrated systems remain largely conceptual, similar hybrid manufacturing architectures combining deposition and in-situ consolidation have been proposed in terrestrial and space-oriented AM research (Vasudeva et al. 2026; Taghizadeh and Zhu 2024).

4.9.2 Resource and logistics

A full CSP-based construction operation on the Moon or Mars would require a complete supply chain, including power generation, regolith excavation, powder preparation, solvent management, pressing, and part finishing. Each link is a challenge. For example, constructing large habitats might require a fleet of CSP units and robotic carriers (Nair et al. 2024). The overall cost-effectiveness must be assessed against alternatives (like cast regolith, geopolimer, or imported materials). Early missions may only use CSP for niche parts or demonstrations until its reliability and benefits are proven. Comparable concepts for robotic construction fleets, autonomous process control, and AI-driven orchestration of multi-robot systems have been proposed for lunar infrastructure and are in development for terrestrial and orbital applications (Rahimi Nohooji and Voos 2025).

Taken together, these challenges highlight that the feasibility of CSP in space depends not on a single limiting factor, but on coordinated solutions spanning materials chemistry, mechanical design, and system-level integration. The key general challenges of applying CSP in space, along with potential mitigation strategies, are summarized in Table 4. Despite these challenges, the early results and engineering analyses are encouraging. Laboratory demonstrations of CSP in simulated space environments have shown that the core process can work. What remains is to develop engineering

Table 4 Key general challenges for CSP in space and potential technological solutions

Challenge	Description	Possible solutions
Solvent availability	Limited water in lunar vacuum	Closed-loop recycling, use of ionic liquids, extraction from ice deposits
High pressure requirement	CSP needs 100–300 MPa	Lightweight mechanical presses, modular compaction systems, energy recovery
Powder preparation	Coarse regolith feedstock	On-site grinding, sieving, beneficiation
Thermal control in vacuum	Heat loss and solvent evaporation	Sealed chambers, low-vapor solvents, insulated CSP modules
Microgravity processing	Uneven liquid distribution	Vibration-assisted mixing, AI feedback control

solutions for these challenges. For instance, the press could be modular (forming bricks one at a time) or integrated into a mobile robot. Solvent loops could recycle all water to high purity. The AI algorithms could adjust pressure and heat in real time to accommodate variability in the soil. To consolidate the discussion, Table 5 summarizes CSP's main opportunities and technical challenges across lunar, Martian, orbital, and asteroid environments, including solvent options, gravity effects, and engineering requirements for each case. In the next section, we discuss how CSP might integrate with such technologies to mitigate its challenges.

5 Integration with emerging technologies

This section focuses on emerging technologies that can mitigate the challenges outlined in Sect. 4 and enable scalable, autonomous CSP-based construction systems. The CSP does not exist in isolation; its future in space will likely be intertwined with advances in robotics, AM, and digital design. The broader vision of integrating CSP with AM, robotics, and AI-driven autonomous construction is summarized in Fig. 5. We highlight several avenues of technological integration:

5.1 Additive manufacturing synergy

The CSP complements 3D printing by allowing each printed layer or slurry to be hardened without high-temperature

Table 5 Summary of CSP opportunities & technical challenges across space environments

Environment	Typical feedstock (composition)	Solvent/binder options	Gravity & operational implications	Pressure requirement (typical)	Processing temperature (typical)
Moon (surface)	Basaltic regolith, glassy agglutinates, ilmenite; dry except polar volatiles	Water (from polar ice), alkaline solutions (NaOH/KOH), low-vapor solvents; sealed dies required	0.16 g, less natural compaction, vacuum causes rapid solvent evaporation	100–300 MPa (lab demos use high uniaxial pressure)	100–200 °C (lab CSP demos ~ 150 °C)
Mars (surface)	Basaltic soil mixed with clays, salts (perchlorates), variable iron content	Water (ice/adsorbed), alkaline activators; careful perchlorate handling or pre-treatment	0.38 g, partial compaction helps; CO ₂ atmosphere eases containment vs. vacuum	30–250 MPa (reported lab ranges; process dependent)	100–250 °C (chemistry-dependent; alkaline aids densification)
Orbital/Cislunar (micro-gravity/pressurized cabins)	Imported regolith samples, recycled spacecraft ceramics/polymers	Aqueous solvents (in-hab water), polymer binders, ionic liquids (low-vapor)	Microgravity: no self-weight, external pressing mandatory; closed environment simplifies vapor containment	100–300 MPa (external press must supply all compaction)	100–250 °C (settable in habitat)
Small bodies/Asteroids (microgravity)	Primitive regolith, possibly carbonaceous or metal-rich fines, water/ice in some bodies	Water from ice, alcohols, deep eutectic solvents, ionic liquids for non-oxide material	Milligravity, anchoring and press reaction forces are required; minimal gravity, easy to move large objects	100–300 MPa (anchored or reaction-force designs)	variable (aqueous CSP ~ 100–250 °C; alternate solvents may require different T)
Environment	Key advantages		Main challenges & engineering requirements		Recommended validation/milestones
Moon (surface)	Low energy vs. conventional sintering; demonstrated high compressive strength (lab lunar bricks)		Sealed processing to avoid solvent loss; solvent extraction/recycling; powder comminution; thermal cycling resilience		Lab vacuum chamber CSP tests; validate solvent recovery in vacuum; thermal cycling & micrometeoroid exposure tests
Mars (surface)	Water available in many locations; regolith chemistry can aid alkali activation; useful structural strength (tens of MPa possible)		Perchlorate reactivity/toxicity; dust/storm tolerance; robust insulation; feedstock variability requiring recipe adaptation		Demonstrate CSP on Mars simulants with perchlorates; solvent recycling and perchlorate mitigation tests; environmental dust tolerance
Orbital/Cislunar (microgravity/pressurized cabins)	On-demand parts fabrication; reuse of waste materials; controlled environment simplifies handling		Press integration in ISS/Gateway compartments; safe solvent handling (toxicity & venting); mass/power constraints		ISS demonstration: sealed CSP press, solvent recovery, automated feed; validate closed-loop control
Small bodies/Asteroids (microgravity)	Potential local volatiles as solvents; access to metal-rich feedstocks for functional parts		Anchoring & reaction force design; solvent selection for non-oxide feedstocks; extreme thermal environment		Demonstrate CSP with non-aqueous solvents on meteorite/analogue material; test anchoring/press designs in parabolic or microgravity platforms

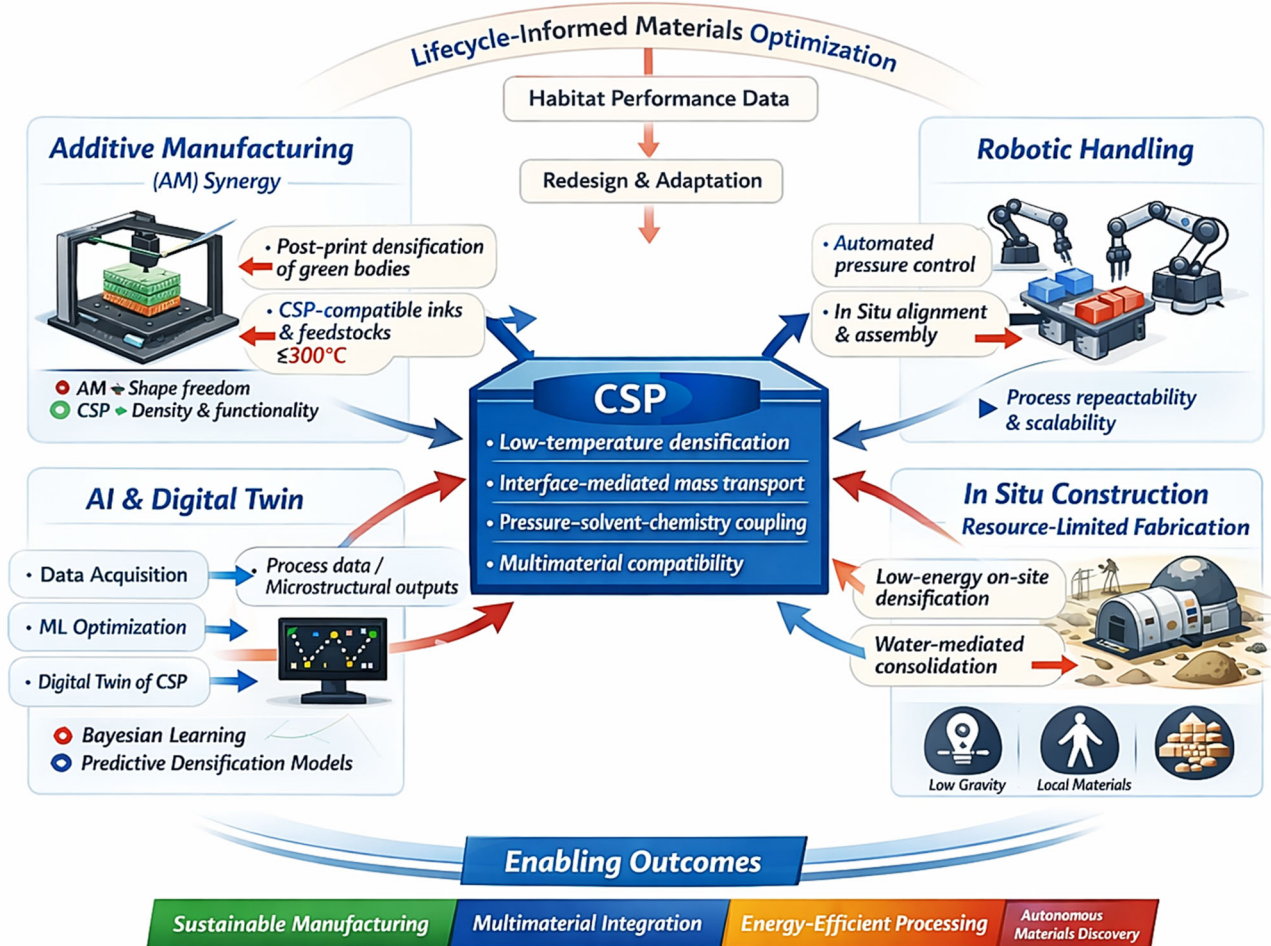


Fig. 5 Integration of CSP with emerging technologies in space exploration. CSP can synergize with AM, robotic construction, and AI-driven process control to enable autonomous fabrication of large-scale, multifunctional habitats

furnaces. For example, a large printer could deposit a regolith–binder paste in layers, then a press or roller could compact and cold-sinter each layer sequentially (Jakus et al. 2017; Osio-Norgaard et al. 2021). This hybrid approach could create fully dense, geometrically complex structures. In practice, ESA’s D-Shape printer concept (for lunar concrete) shows that binder-jetting of regolith is feasible (Isachenkov et al. 2021); CSP could act as the post-process densifier. Moreover, CSP can enable multimaterial printing: by combining ceramic powders with polymer or metal powders (Sun et al. 2025), the printed part can be consolidated into a composite with tailored properties in one sintering step (Lai et al. 2025). This opens the door to printing smart panels (e.g., embedding copper tracks for electronics) or reinforcing fibers into walls.

5.2 Artificial intelligence and digital process control

The CSP parameter space (pressure, temperature, solvent chemistry, and time (Ndayishimiye et al. 2020a)) is complex, especially when materials vary. Machine learning and AI can help optimize these parameters quickly. For instance, an onboard AI could use sensors (e.g. densification gauges) to adjust pressure in real time, achieving uniform density even as regolith composition drifts (Kurek et al. 2025; Peña-Asensio et al. 2024). Similarly, generative design algorithms could plan the shape and infill pattern of cold-sintered parts to meet structural and thermal requirements. For example, AI-driven topology optimization might produce a sandwich panel where inner pores are left for insulation while outer shells are fully dense for strength, all feasible by controlling the CSP process locally. Digital twins of the CSP system (using computational models of pressure-solution sintering) could simulate new recipes before they are tried, reducing trial-and-error (Niu et al. 2024). In short, AI offers to make

CSP adaptive and efficient in the variable, unforgiving conditions of space. Practically, closed-loop control should be implemented with a small sensor suite (acoustic emission for neck formation, dielectric/impedance probes for liquid content, and real-time displacement/density gauges) coupled to onboard ML models, enabling adaptive pressure/temperature profiles when feedstock composition varies.

To move beyond a conceptual role, the application of AI in CSP can be explicitly framed in terms of defined input–output relationships and operational scenarios. In AI-assisted CSP optimization, input parameters may include applied pressure, sintering temperature, dwell time, solvent chemistry and concentration, particle size distribution, and regolith mineralogical composition (Liu et al. 2024; Ginés-Palomares et al. 2023). The corresponding output parameters of interest include relative density, porosity distribution, compressive strength, dimensional accuracy, and process energy efficiency. Training data for such AI models can be sourced from a combination of terrestrial CSP experiments reported in the literature, targeted high-throughput CSP studies using lunar and Martian regolith simulants, and physics-informed synthetic datasets generated from mechanistic dissolution–precipitation models. In future in-space deployments, these datasets can be continuously updated using real-time sensor data, such as force–displacement responses, temperature evolution, and densification kinetics, enabling adaptive learning and on-orbit model refinement (Maier 2024). Within this framework, AI can support several specific application scenarios for in-space CSP: (i) real-time optimization of pressure–temperature schedules under variable power availability; (ii) adaptive compensation for feedstock variability and impurity-induced process fluctuations; and (iii) closed-loop quality control for autonomous CSP units operating with minimal crew involvement (Nie et al. 2023). These scenarios highlight AI not merely as a post-processing tool, but as an active decision-making layer that enhances robustness, reliability, and autonomy in CSP-based in-space manufacturing systems.

5.3 Microstructure design strategies for enhanced mechanical performance

Controlling microstructure evolution is central to improving the mechanical performance of materials fabricated via the CSP (Lai et al. 2024b). Unlike conventional sintering, CSP enables densification at low temperatures, preserving fine grain sizes, metastable phases, and engineered interfaces that would otherwise coarsen during high-temperature processing (Ndayishimiye et al. 2020a). Several parameters govern microstructural design in CSP-fabricated materials, including particle size distribution, surface chemistry, transient liquid composition, applied pressure, and dwell time (Rahimi Mamaghani and Parvin 2025a). For instance, As

the primary particle size decreases into the nanometer range, powder agglomeration becomes a critical factor that can hinder reliable powder processing (Nur et al. 2021). The integration of CSP with AM, spark plasma-assisted pressing, or external field-assisted techniques further enables spatial control of densification and microstructural gradients (Rahimi Mamaghani et al. 2025b). Such hybrid approaches allow the fabrication of architected materials with tailored porosity, anisotropic grain alignment, or reinforced interphases, directly impacting fracture toughness, strength, and fatigue resistance (Singh et al. 2024). Additionally, interface engineering, through dopant segregation, core–shell particles, or functional surface coatings, offers a powerful route to enhance mechanical reliability without increasing processing temperature. These strategies highlight CSP not merely as a low-temperature densification method, but as a microstructure-by-design platform for next-generation structural and functional materials.

5.4 Space architecture and habitat design

Cold-sintered materials will shape the design of space structures. Architects and engineers must account for CSP's capabilities, such as high compressive strength but limited tensile strength (Rahimi Mamaghani and Parvin 2025b). A habitat wall built from CSP bricks might use an arch or vault geometry to exploit compression. The uniformity of CSP bricks can reduce the need for mortar joints. Furthermore, CSP enables functionally graded materials; by varying binder content or particle mix during sintering, one could create walls with a gradient in thermal conductivity (high insulation inside, high thermal mass outside) or integrate radiation shields where needed. The concept of smart bricks also arises; for example, embedding fiber-optic or fiber-Bragg-grating sensors within the powder so that after sintering, the brick can monitor strain or temperature over time, as studied in recent ISRU smart-brick research (Souza et al. 2024).

5.5 Radiation shielding and multifunctional use

Combining CSP with materials science, one can enhance radiation protection. Certain additives (like boron-rich glasses or hydrogenous polymers) could be co-sintered with regolith to create composite shields that attenuate both gamma rays and neutrons (Alipoor and Eshghi 2025). Lightweight regolith-based metamaterials might be designed via CSP to achieve specific attenuation spectra. Even the surface topology (such as dimpled or layered bricks) can be designed to scatter micrometeoroids or trap secondary radiation. In effect, CSP allows the fabrication of purpose-built shielding blocks customized to the mission's radiation profile, which is a space-architectural design task.

5.6 Sustainability and life support integration

Since CSP uses water/solvents, integrating life support is important. For example, water recovered from habitat humidity could be purified and reused as the sintering liquid, linking CSP with the Environmental Control and Life Support System (ECLSS) (Adu et al. 2025). Conversely, some CSP binders might be derived from bioreactors, such as bacterial production of polysaccharides as binders. Such cross-discipline integration means that CSP units could become part of a larger closed-loop ecosystem on long-duration missions.

In essence, CSP is poised to be a holistic building block technology. Its success will depend not only on materials science but also on robotics (for automated feeding and pressing), software (for process control), and design innovation (to exploit what cold-sintered materials can do). A conceptual system integrating excavation, powder preparation, CSP pressing, and robotic assembly into habitat construction is presented in Fig. 6. Emerging space construction projects should consider CSP alongside alternative methods, integrating it where its strengths (low-energy, ISRU compatibility, composite fabrication) best serve mission needs.

6 Future outlook and research directions

The CSP for space applications is an emerging field, and many avenues remain to explore. We outline key future directions:

6.1 Validation in space conditions

Next steps include testing CSP under more realistic conditions. This could start with vacuum chamber experiments and parabolic flight tests to study microgravity effects on the solvated flow and densification. Long-term vacuum exposure tests will show how CSP products age in cosmic ray environments. Demonstration projects (e.g., a CSP press on the ISS or a lunar lander) would provide invaluable data.

6.2 Alternative solvents and binders

Research should examine non-aqueous and low-volatility binders suitable for space. For example, formamide, glycerol, liquid CO₂, or ionic liquids might serve as transient phases without boiling off. Additionally, exploring regolith-intrinsic binders (like using native sulfates or perchlorates activated by heat) could eliminate external chemicals (Sun et al. 2025; Geng et al. 2023). The goal is to minimize consumables brought from Earth. Table 6 lists candidate solvents for CSP under extraterrestrial conditions, highlighting their key properties, limitations, and possible in-situ sourcing strategies.

6.3 Powder preparation and feedstock blends

Work is needed on efficient comminution or beneficiation of regolith into CSP-friendly powders. Ball milling regolith dust or using fluidized-bed micronizers could be tested in lunar simulators. Researchers should also study composites of regolith with other local materials (like charcoals, metals from asteroid ore, or polymeric waste) to create designer mixtures. For instance, adding shredded plastic could reduce brittleness, making a regolith–polymer hybrid through CSP.

6.4 Process modeling and control

Advanced models of CSP that account for multicomponent mixtures under low-gravity conditions should be developed. These models would enable predictive control of porosity and strength (Wang et al. 2020). Coupling thermal, chemical, and mechanical simulations, possibly using machine learning calibrated with experiments, could accelerate CSP process optimization. In parallel, in-situ monitoring techniques, such as acoustic or impedance sensing during sintering, could be integrated to give real-time feedback (Maier 2024).

6.5 Large-scale automation

Moving from bricks to buildings will require new equipment. Ideas include a continuous CSP conveyance, like a hot isostatic press on a conveyor belt, or robotic bricklaying where a robot places CSP-formed blocks and then sinters them in place (Nair et al. 2024). Developing lighter, airless presses (perhaps using mechanical advantage systems or smart materials) is crucial. Partnerships with the growing space robotics community (e.g. NASA's Lunar Gateway projects or commercial lunar landers) could drive prototype development.

6.6 Multi-functional structures

Beyond structural uses, CSP materials could serve as sensors, batteries, or life-support components. For example, sintering regolith with embedded salt might create humidity or gas sensors that respond to environmental changes (Zhou et al. 2024; Han et al. 2022). Sintered ceramics could hold catalysts for air/water recycling or even capture volatiles from the regolith itself. Future research should explore these cross-functional possibilities, turning building materials into living parts of a space habitat's system.

6.7 Integration with other ISRU methods

The CSP will likely coexist with methods like sintering by microwaves or solar concentrators, and with chemical processing (like extracting metals). Hybrid schemes, such as partially melting a CSP block with microwaves to seal it,

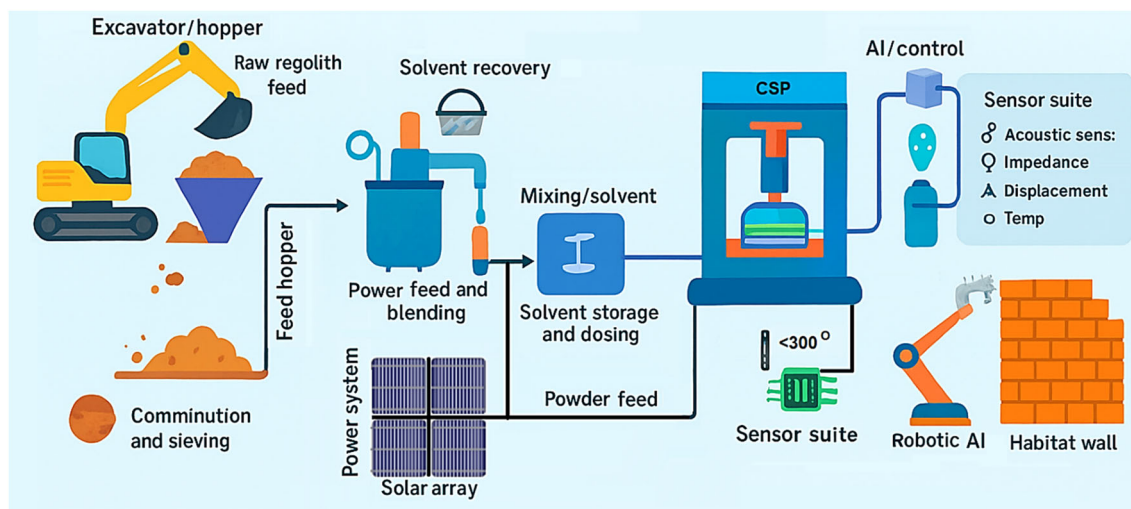


Fig. 6 Conceptual vision of a robotic CSP-based construction system. Regolith is excavated and milled into powder, fed into a compact press with solvent recycling, and assembled robotically into bricks or layered

walls. Integration with AM and AI-driven process control is shown schematically

Table 6 Candidate solvents for CSP in space applications

Solvent/Binder	Volatility/Stability	Advantages	Challenges	Possible in-situ source
Water (H ₂ O)	High volatility in vacuum	Universal solvent, well-studied	Scarcity on Moon; needs recycling	Lunar ice, Martian subsurface ice
NaOH/KOH solution	Moderate volatility	Enhances dissolution of silicates	Corrosive, toxic, requires containment	Electrolysis of salts in regolith (Mars)
Ionic liquids	Very low volatility	Stable under vacuum, recyclable	Heavy to transport, limited ISRU pathways	Import from Earth initially
Formamide	Low volatility	Good solvent for oxides, stable	Toxic, requires handling care	Possible organic synthesis in-situ
Glycerol	Low volatility	Non-toxic, reusable	Less effective at dissolving some minerals	Bioreactors with algae or bacteria
Liquid CO ₂ (supercritical)	Medium volatility	Abundant on Mars, tunable solvent	Requires high-pressure vessels	Martian atmosphere
Ammonia/ammonium salts	Moderate volatility	Dual role: solvent and fertilizer precursor	Hazardous, needs containment	Found in cometary ice, asteroid volatiles

may yield even stronger products (Zhang et al. 2024c). Coordinated studies comparing CSP with hot-sintering, polymer binding, and casting methods will clarify where each technique excels (Li et al. 2025b). In addition, CSP could be combined with advanced cold-sintering routes such as flash sintering (Dang et al. 2025) and spark plasma sintering (SPS) (Han et al. 2024), techniques that are also under active investigation for potential applications in space. In the next decade, we foresee CSP research moving from laboratory proofs to field validation. As NASA and other agencies plan Artemis and Mars missions, CSP could be included in roadmaps for ISRU demonstration, for example, building a small demo hut of cold-sintered bricks on the Moon. Collaborations between material scientists, space engineers, and architects

will be essential to fully realize CSP's potential. We propose a staged technology readiness levels (TRL) roadmap for CSP in space: (i) TRL 3–4: vacuum-chamber and thermal cycling tests plus parabolic flights; (ii) TRL 5–6: ISS or suborbital sealed-chamber demonstrations; (iii) TRL 7–8: robotic lander demonstration producing a modest mock-habitat element on the lunar surface as an Artemis payload.

To move beyond a conceptual roadmap, the technology readiness progression of CSP for space applications can be defined through task-oriented milestones and acceptance criteria aligned with standard frameworks for assessing technology maturity. At TRL 1–2, feasibility is established through laboratory-scale CSP densification of representative

regolith simulants, demonstrating pressure-assisted dissolution–precipitation at temperatures below 300 °C. TRL 3 focuses on proof-of-concept validation, requiring reproducible mechanical performance and parameter–property correlations under controlled terrestrial conditions. Experimental studies demonstrating this, such as those achieving compressive strength up to 193.88 MPa with lunar regolith simulants, form the foundational research for this stage (Xue and Qiao 2025b). At TRL 4, CSP is extended to component-level fabrication, with acceptance based on geometric fidelity and stability under simulated vacuum and thermal cycling. TRL 5 requires validation in relevant environments, such as reduced-pressure or partial-gravity analogs, with demonstrated solvent containment and pressure control. This progression directly supports key capability areas for lunar exploration, such as ISRU and excavation and construction. TRL 6 involves system-level integration of CSP with robotic handling, AM, or AI-assisted process control, culminating in autonomous or semi-autonomous fabrication demonstrations. TRL 7–8 are achieved through in-space or surface-based demonstrations, where CSP-produced elements meet structural or shielding performance requirements under operational conditions. Finally, TRL 9 corresponds to sustained, mission-relevant deployment of CSP-enabled construction using in-situ resources, with verified reliability and repeatability. This ultimate stage fulfills the goal of establishing a robust, ISRU-based manufacturing capability for sustained lunar exploration and future Mars missions. In conclusion, CSP represents a promising tool for off-world construction and manufacturing. By dramatically lowering the energy and temperature needed to densify materials, CSP opens up new possibilities for building with extraterrestrial soils. While significant engineering challenges remain, ongoing advances in materials science and space technology suggest that CSP could become a cornerstone of future space habitats, infrastructure, and exploration equipment. Its ability to transform raw regolith into high-performance ceramics could, in the coming years, help humankind build beyond Earth in a sustainable and resilient way.

7 Conclusion

Cold sintering offers a practical, lower-energy pathway to convert local soils into structural and multifunctional elements, complementing mature ISRU technologies and enabling mission-level benefits. By enabling high-strength consolidation of local regolith and composites at drastically reduced temperatures and energy costs, CSP addresses the most critical barriers to extraterrestrial construction: power limitations, material transport, and adaptability to harsh environments. Recent demonstrations on lunar and Martian simulants confirm that CSP can produce dense, durable

bricks and components suitable for habitats, shielding, and infrastructure. Its compatibility with polymers, metals, and functional additives extends its utility beyond simple construction to advanced systems such as sensors, energy devices, and multifunctional walls. The path forward requires resolving practical challenges, designing lightweight high-pressure presses, developing closed-loop solvent cycles, and validating performance under vacuum, radiation, and microgravity. Integration with AM, robotics, and AI-driven process control will enable CSP to evolve into a fully autonomous construction technology. In parallel, space architects must explore new structural concepts tailored to the compressive strengths and graded properties CSP materials offer. Looking ahead, CSP is more than a fabrication method: it is a strategy for embedding sustainability and adaptability into space exploration. Whether on the Moon, Mars, or in deep space, CSP could empower astronauts and robotic systems to transform local soils into the foundations of human presence. In this sense, CSP stands not only as a scientific advance but as an enabling technology for the next chapter of human civilization, building resilient worlds beyond Earth.

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Data availability No data was used for the research described in the article.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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